

WEEK 13 to 14: ORGAN SYSTEM

SKELETAL SYSTEM

- Rigid internal framework for support, protection, movement, mineral storage, and hematopoiesis

FUNCTIONS

1.) Support

- Structural framework
- Maintains posture and supports soft tissues
- Attachment point for muscles and tendons

2.) Protection

- Shields vital organs:
 - Skull - brain
 - Vertebral column - spinal cord
 - Thoracic cage - heart and lungs
 - Pelvis - reproductive and urinary organs

3.) Movement

- Bones as levers, joints as fulcrums, and muscles provide force
- Type and ROM depend on:
 - Joint structure, muscle attachment, length and orientation of lever arms, long bones (e.g., femur, humerus) act as levers for locomotion, and short bones (e.g., carpals, tarsals) allow stability and fine motion

4.) Mineral storage

- Reservoir for Ca and P for:

- Muscle contraction
- Nerve transmission
- Enzyme function
- Blood clotting

- Hormonal regulation

- **PTH** increases blood Ca by stimulating osteoclast activity
- **Calcitonin** decreases blood Ca by promoting Ca deposition in bone

5.) Blood cell formation — “hematopoiesis”

- Occurs in red bone marrow in:
 - Flat bones (sternum, ribs, skull, pelvis)
 - Proximal epiphyses of long bones (femur, humerus)
- RBC, WBC, and platelet production

6.) Energy storage

- Yellow bone marrow store triglycerides (fat) and nutrients for energy reserve

- With aging, red bone marrow converts to yellow bone marrow

COMPOSITION

1.) Bones — “osseous tissue”

- Dense connective tissue consisting of cells, fibers, and a mineral matrix

COMPONENTS

- 1.) **Osteoblast** - bone-forming cells; secretes *osteoid* (organic portion of bone matrix)
- 2.) **Osteocytes** - mature bone cells in lacunae; maintain bone matrix
- 3.) **Osteoclasts** - resorb bone during remodeling; large, multinucleated cells; remove or renew old bones

COMPOSITION

- 1.) **Organic (35%)** - type I collagen for tensile strength and flexibility
- 2.) **Inorganic (65%)** - hydroxyapatite crystals ($\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$) for hardness and rigidity; absorbs impact

CHARACTERISTICS

- 1.) Highly vascular
 - 2.) Capable of remodeling (continuous resorption and deposition)
 - 3.) Sensitive to mechanical loading (*Wolff's Law*)
- bone adapts to stress by becoming stronger

2.) Cartilage

- Specialized, semi-rigid connective tissue consisting of glycosaminoglycans and proteoglycan-rich ground substance

- Attracts water for high degree of hydration

- Avascular; low regeneration and repair

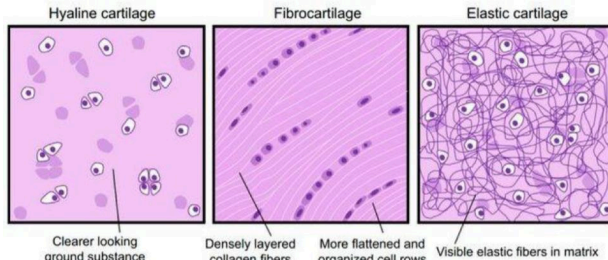
TYPES

- 1.) **Hyaline cartilage** - most common; found at articular surfaces, costal cartilages, trachea, larynx, and nasal septum; glassy extracellular matrix; has smooth surface and for structural support
- 2.) **Elastic cartilage** - flexible; found in external

WEEK 13 to 14: ORGAN SYSTEM

ear (pinna), epiglottis, Eustachian tube

3.) Fibrocartilage - tough, shock-absorbing, resistant; found in intervertebral discs, pubic symphysis, menisci



Cartilage type	ECM	Cells	Organization of cells	Perichondrium	Locations / Functions
Hyaline cartilage	Type II collagen, abundant aggrecan, & various other proteoglycans & GAGs	Chondroblasts, chondrocytes, & fibroblasts in perichondrium	Isolated in lacunae or in small isogenous groups	Yes, contains fibroblasts, mesenchymal stem cells, blood vessels, & small nerves	Aids in sliding within joints, serves as a model for bone growth, & provides structural support to airway walls
Elastic cartilage	Type II collagen & elastic fibers, aggrecan & various other proteoglycans & GAGs	Chondroblasts, chondrocytes, & fibroblasts in perichondrium	In lacunae as small isogenous groups	Yes, contains fibroblasts, mesenchymal stem cells, blood vessels, & small nerves	Provides flexible shape and support to soft tissues such as the external ear, auditory tube, & epiglottis
Fibrocartilage	Type I & type II collagen & aggrecan	Chondrocytes & fibroblasts	Isolated in lacunae and sometimes in small groups aligned in rows	No	Provides cushioning, strength, and resistance to tearing and compression within intervertebral discs, meniscus, and pubic symphysis

FUNCTION

- 1.) Reduces friction within bones
- 2.) Model for bone formation (endochondral ossification)
- 3.) Provides structural support with limited repair capacity (avascular)

3.) Ligaments

- Dense regular connective tissue composed of collagen fibers, which connect bone to bone across joints
- Passive stabilizers for joint integrity while allowing controlled motion
 - Acts as restraints

COMPOSITION

- 1.) **Cells** - fibroblasts for synthesizing collagen and extracellular matrix
- 2.) **Fibers** - type I collagen arranged in parallel bundles for high tensile strength
- 3.) **Ground substance** - minimal; proteoglycans for resisting compressive forces
- 4.) **Vascular supply** - poor; slow healing after injury
- 5.) **Innervation** - proprioceptive (repetitive ankle

sprain: ACL tear) and nociceptive nerve endings for joint positioning feedback and pain detection

FUNCTION

- 1.) Stabilize joints - ACL: prevents overextending of tibia from the femur; MCL: prevent excessive internal forces
- 2.) Restrict excessive movement and maintain alignment

MECHANICAL PROPERTIES

1.) Viscoelastic behavior

- Deforms under stress and gradually return to original shape
- Important for absorbing energy during sudden movements

2.) Creep and stress relaxation

- Prolonged loading cause gradual elongation, contributes to joint laxity if overstretched and unable to recoil

3.) Failure point

- Occurs when fibers exceed their elastic limit (between 6-8% strain)

4.) Joints — “articulations”

- Point of connection for two or more bones for movement and mechanical support to the skeleton

STRUCTURAL AND FUNCTIONAL CLASSIFICATION

1.) Fibrous joints

- Bones are connected by dense fibrous connective tissue; no joint cavity
- Mobility: immovable (synarthrosis)
- Examples: sutures of the skull, distal tibiofibular joint

2.) Cartilaginous joints

- Bones are connected by cartilage (hyaline or fibrocartilage); no joint cavity
- Mobility: slightly movable (amphiarthrosis)
- Examples: Intervertebral discs (symphysis - joined by fibrocartilage), epiphyseal plate (synchondrosis - joined by hyaline cartilage)

3.) Synovial joints

- Bones are separated by a fluid-filled joint cavity

WEEK 13 to 14: ORGAN SYSTEM

and enclosed by a capsule; for lubrication and reducing friction

- Mobility: freely movable (diarthrosis)
- Examples: shoulder, hip, knee (tibiofemoral), elbow (humeroulnar)

FEATURES OF SYNOVIAL JOINTS

- Most complex and mobile type of joint; allows wide ROM

1.) Articular cartilage

- Smooth hyaline cartilage covering the articulating bone surfaces; ends of bones
- Avascular
- Friction reduction and shock absorption

2.) Joint (articular) capsule

- Encloses joint; composed of:
 - Fibrous capsule - outer tough layer for stability
 - Synovial membrane - lines joint cavity; inner layer producing synovial fluid

3.) Synovial fluid

- Viscous, lubricating fluid; reduces friction, nourishes avascular cartilage, and acts as a shock absorber

4.) Joint cavity

- Space between the articulating bones that contains synovial fluid

5.) Accessory structures:

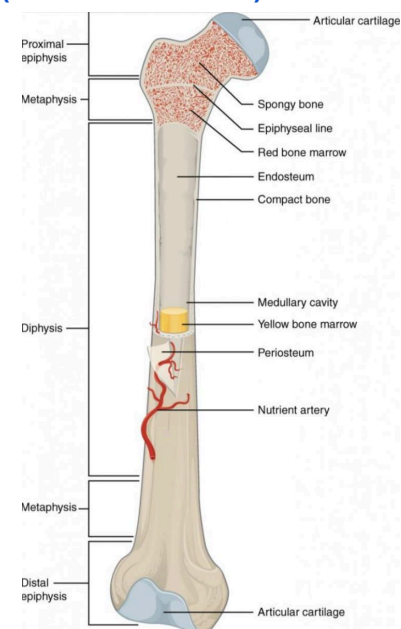
- Ligaments - bone to bone; reinforce joint and limit excessive motion
- Tendons - muscles to bones; aid in joint stability
- Bursae - Fluid-filled sacs that reduce friction between bone, tendon, and skin
- Menisci (Articular discs) - fibrocartilaginous pads that improve fit between articulating surfaces and absorb shock in weight-bearing joints; only located knee

- Long bones (e.g., femur, humerus)
 - Act as levers
- Short bones (e.g., carpals, tarsals)
 - Provide stability with limited motion
- Flat bones (e.g., skull, ribs, scapula)
 - Protect organs and allow muscle attachment
- Irregular bones (e.g., vertebrae, pelvis)
 - Complex shapes
- Sesamoid bones (e.g., patella, pisiform - smallest)
 - Develop in tendons to reduce friction

2.) By location

- Axial skeleton (skull, vertebral column, ribs, sternum)
- Appendicular skeleton (upper and lower limbs, pectoral and pelvic girdles)

GROSS (MACROSCOPIC) STRUCTURE



1.) Compact bone — “cortical bone”

- Dense, hard outer layer of the bone; forms the shaft (diaphysis) and outer shell of all bones
- **Organized into osteons (Haversian systems)** - cylindrical & functional unit containing concentric lamellae around a central canal (contains BVs)
- **Function:**
 - Strength, rigidity, and protection

CLASSIFICATION

1.) By shape

WEEK 13 to 14: ORGAN SYSTEM

- Supports weight and resists bending

2.) Spongy bone — “cancellous or trabecular bone”

- Porous, lighter inner layer of bone found in the epiphyses (end) and inside flat bones
- Composed of a network of trabeculae (bony plates) with spaces filled with red bone marrow
- **Function:**
 - Provides strength without excessive weight
 - Site of hematopoiesis

3.) Periosteum

- Tough, outermost, fibrous membrane; covers the outer surface of bone, except at joint articular surfaces
- **Layers:**
 - Fibrous layer - outer dense connective tissue; *site for muscle, tendon, and ligament attachment*
 - Osteogenic layer - Inner layer with osteoblasts for *bone growth and repair*
- **Function:**
 - Protects bone
 - Nourishes bone tissue (contains BV)
 - Anchors muscles and ligaments
 - Involved in bone growth and healing

4.) Endosteum

- Innermost; Thin membrane lining the inner surfaces of the medullary cavity and trabeculae of spongy bone
- **Function:**
 - Contains osteoblasts and osteoclasts
 - Vital role in bone remodeling and repair

5.) Medullary (marrow) cavity

- The central cavity of the diaphysis (shaft) of long bones
- **Contents:**
 - Red bone marrow in children (active in blood formation)
 - Yellow bone marrow in adults (fat storage)
- **Function:**

- Storage site for marrow and energy reserves
- Important for hematopoiesis in young bones

6.) Articular cartilage

- Smooth layer of hyaline cartilage; covers the ends of bones at joints
- **Function:**
 - Friction reduction and shock absorption

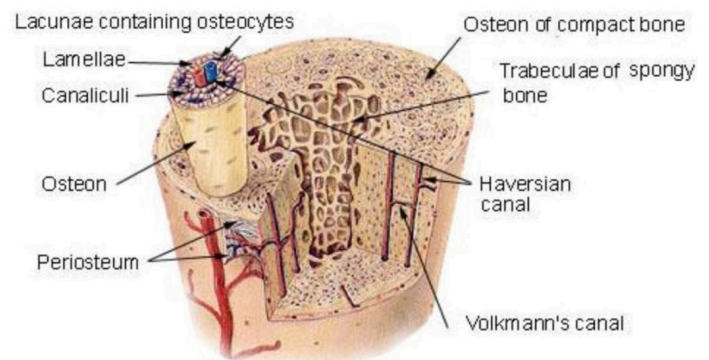
7.) Nutrient foramina

- Small opening (holes) on the bone surface
- **Function:**
 - Allow entry of BVs and nerves into the bone
 - Provide nutrition and innervation to bone and marrow

8.) Metaphysis

- Contains growth plate; between epiphysis and diaphysis

MICROSCOPIC STRUCTURE



1.) Osteon — “Haversian system”

- **Structural unit of compact bone**; consist of concentric lamellae surrounding central canal
- **Function:** strength and efficient nutrient delivery

2.) Haversian canal

- Central channel in osteon containing BVs and nerves
- **Function:** nourishes bone cells

3.) Volkmann's canal

WEEK 13 to 14: ORGAN SYSTEM

- Perforating canals connecting adjacent Haversian canals
- Function: communication and nutrient exchange

4.) Lamellae

- Concentric rings of calcified matrix
- Function: strength and resilience

5.) Lacunae

- Small spaces between lamellae containing osteocytes
- Function: house mature bone cells

6.) Canaliculi

- Tiny channels connecting lacunae
- Function: exchange of nutrients and wastes between osteocytes

TYPES OF BONE MARROW

1.) Red bone marrow

- Rich in hematopoietic cells
- Location: flat bones (sternum, ribs, pelvis) and epiphyses of long bones
- Function: produces RBCs, WBCs, and platelets

2.) Yellow bone marrow

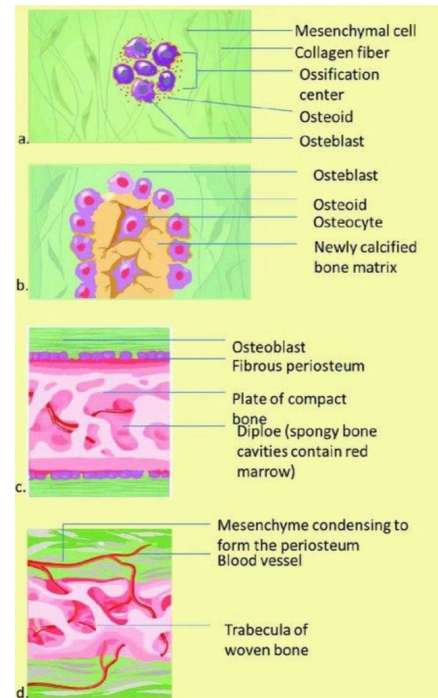
- Mainly in adipose tissue
- Location: medullary cavity of long bones
- Function: stores fat for energy reserve

BONE DEVELOPMENT AND GROWTH

- **Osteogenesis** - process of forming new bone tissue

TYPES OF OSSIFICATION

1.) Intramembranous ossification



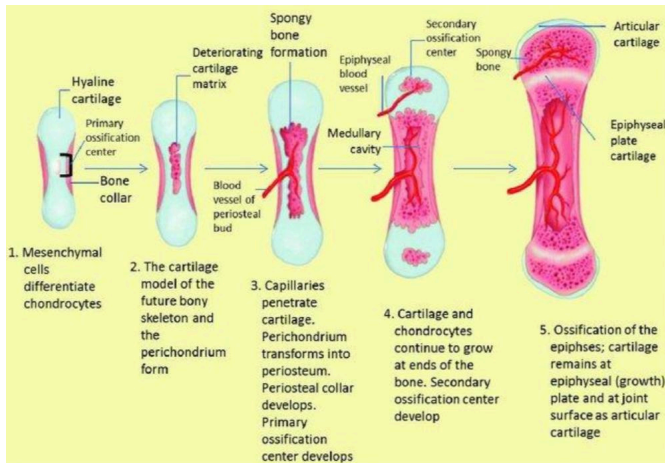
- Formation occurs from mesenchymal (embryonic connective) tissue, without cartilage model
- Examples: flat bones of the skulls (frontal, occipital, parietal), mandible, clavicle (partly formed this way)

STEPS

- 1.) Mesenchymal cells cluster and differentiate into osteoblasts, forming an ossification center
- 2.) Osteoblasts secrete osteoid (bone matrix), which later calcifies
- 3.) As osteoblasts become trapped, they mature into osteocytes within lacunae
- 4.) Trabeculae form and fuse to create spongy bone
- 5.) Outer surfaces are remodeled into compact bone, forming a periosteum

2.) Endochondral ossification

WEEK 13 to 14: ORGAN SYSTEM



- Formation occurs by replacing a pre-existing cartilage model (hyaline) with bone tissue
- Bone growth ends: 21 yrs old in male; 18 yrs old in female
- Examples: long bones (femur, humerus, tibia), and short bones (vertebrae)

STEPS

- 1.) Cartilage model forms in the shape of the future bone
- 2.) Chondrocytes in the center enlarge, calcify, and die, leaving cavities
- 3.) BVs invade cavities, bringing osteoblasts and osteoclasts
- 4.) Primary ossification center develops in the diaphysis (shaft)
- 5.) Secondary ossification centers form later in the epiphysis (ends)
- 6.) Epiphyseal plate (growth plate) remains between diaphysis and epiphysis to allow further elongation

BONE GROWTH

1.) Longitudinal growth — lengthwise

- Site: epiphyseal plates between epiphysis and diaphysis
- Process:
 - Chondrocytes in the cartilage proliferate, enlarge, and are replaced by bone tissue through endochondral ossification
 - New cartilage continues to form on the epiphyseal side, while ossification occurs on the diaphyseal side

- Result: bone elongates until epiphyseal plates ossify (close), forming the epiphyseal line (around late adolescence)

2.) Appositional growth — width or diameter growth

- Site: occurs on the outer surface of bone under periosteum
- Process:
 - Osteoblasts in periosteum deposit new osteoid on the outer surface
 - Osteoclasts in endosteum resorb bone on the inner surface, enlarging the medullary cavity

- Result: bone increases in thickness and strength while remaining light

REGULATION OF BONE GROWTH

1.) Growth hormone (GH)

- Secreted by anterior pituitary; stimulates **chondrocyte proliferation** at epiphyseal plates and **osteoblast activity** for longitudinal and appositional growth

2.) Thyroid hormone

- Enhances GH effects; ensures proper bone proportion and metabolism

3.) Sex hormones — estrogen and testosterone

- Promote rapid growth at puberty; later cause epiphyseal plate closure, ending longitudinal growth

4.) Calcitonin & parathyroid hormone (PTH)

- Regulate Ca and P balance for bone remodeling

5.) Vitamin D — calcitriol

- Aids in Ca absorption and mineralization of bone matrix

6.) Mechanical stress — exercise, gravity

- Stimulates osteoblast activity -> bones adapt and become stronger (Wolff's law)

WEEK 13 to 14: ORGAN SYSTEM

CLINICAL RELEVANCE

1.) *Osteoarthritis*

- Caused by degeneration of articular cartilage leading to pain and stiffness

2.) *Dwarfism*

- Caused by defection during endochondral ossification

3.) *Epiphyseal plate damage*

- In childhood, it can cause growth arrest or angular deformities

4.) *Appositional growth*

- Continues through life as part of bone remodeling in response to stress, exercise, or injury

5.) *Gigantism*

- Excess GH before epiphyseal plate closure
- Abnormal bone length

6.) *Acromegaly*

- Excess GH after plate closure
- Thickened bones (especially face, hands, feet)

7.) *Rickets (children) / Osteomalacia (adults)*

- Vitamin D deficiency
- Poor bone mineralization

8.) *Osteoporosis*

- Bone resorption exceeds formation
- Fragile bones

9.) *Fracture healing*

- Involves reactivation of intramembranous and endochondral ossification processes

WEEK 13 to 14: ORGAN SYSTEM

STRUCTURAL ORGANIZATION OF THE SKELETAL SYSTEM

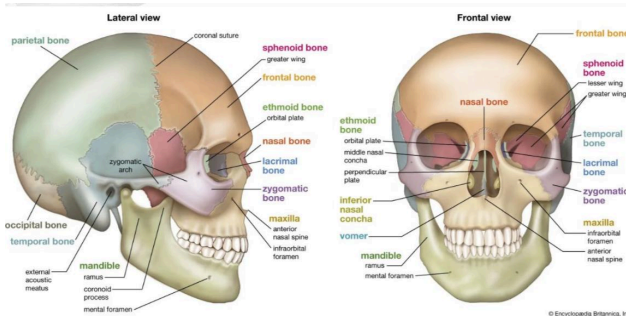
DIVISIONS OF THE SKELETON

1.) Axial skeleton — 80 bones

- Central axis of the body for support and protection of the CNS and vital organs
- Provides structural core from which the appendicular skeleton articulates and moves

COMPONENTS

1.) Skull — 22 bones



a.) Cranial bones — 8 bones

- Function: form the cranial cavity (calvarium) housing and protecting the brain

- **STEP OF:** Sphenoid, Temporal (2), Ethmoid, Parietal (2), Occipital, Frontal

b.) Facial bones — 14 bones

- Function: framework of face; provide cavities for sense organs; anchor facial muscles

- **Vicky Can Not Make My Pet Zebra Laugh:** Vomer, inferior nasal Concha (2), Nasal (2), Maxilla (2), Mandible, Palatine (2), Zygomatic (2), Lacrimal (2)

2.) Hyoid bone

- Function: anchoring structure for the tongue and muscles of swallowing and speech
- Located in anterior neck between mandible and larynx (C3)
- Doesn't articulate directly with other bone; suspended by muscles and ligaments

3.) Vertebral column — 26 bones in adults; 33 bones in children

- Function:
 - Supports the skull and trunk
 - Protects SC

- Provides attachment for ribs and back muscles
- Allows flexibility and motion (flexion, extension, rotation)

a.) Cervical - 7 (C1-C7)

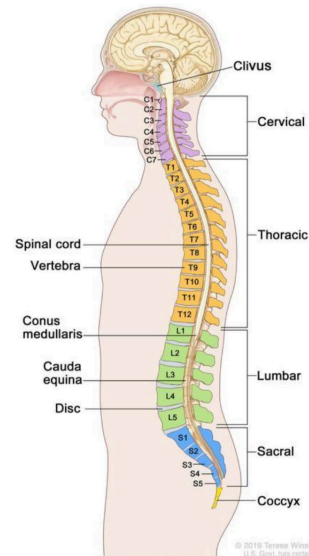
b.) Thoracic - 12 (T1-T12)

c.) Lumbar - 5 (L1-L5)

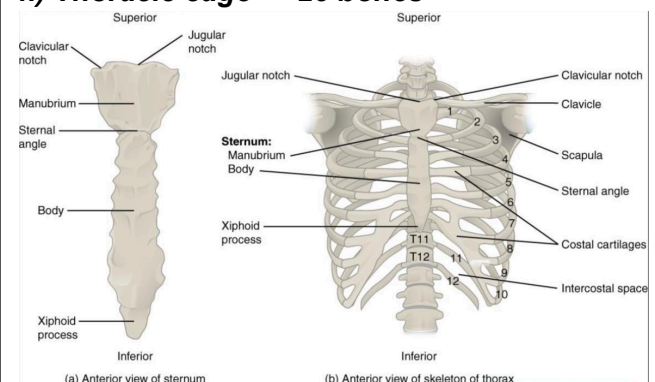
d.) Sacrum - 1 (fusion of 5 sacral vertebrae)

e.) Coccyx - 1 (fusion of 4 small vertebrae)

- Primary curves: thoracic and sacral curves
- Secondary curves: cervical (develops first) and lumbar (9-10 months); C and L develops through walking



4.) Thoracic cage — 25 bones



Function:

- Encloses and protects thoracic organs (lungs, heart)
- Supports pectoral girdle and upper limbs

WEEK 13 to 14: ORGAN SYSTEM

- Aids in respiration by expanding and contracting during breathing

a.) Ribs — 24 bones

- 7 true ribs (attach directly to sternum via costal cartilage)
- 3 false ribs (attach indirectly)
- 2 floating ribs (no anterior attachment)

b.) Sternum — 1 bone

- Manubrium, body, Xiphoid process

5.) Auditory ossicles — 6 bones

- Malleus - 2
- Incus - 2
- Stapes - 2

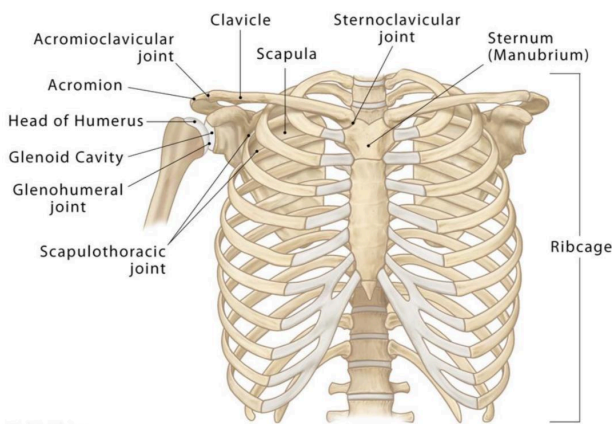
2.) Appendicular skeleton — 126 bones

- Consists of the bones of the limbs and the girdles that attach them to the axial skeleton
- Facilitates movement, locomotion, and manipulation of the environment

COMPONENTS

1.) Pectoral (shoulder) girdle

Pectoral Girdle



- Function:

- Provides mobility and ROM to upper limb
- Muscular attachment platform for shoulder and arm movement

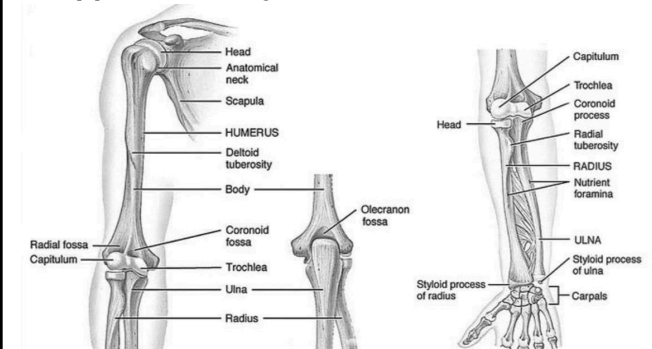
a.) Clavicle — collarbone

- S-shaped bone connecting sternum to scapula
- Strut holding the upper limb away from the trunk; strut bone

b.) Scapula — shoulder blade

- Triangular bone; provides wide surface area for muscle attachment
- Contains glenoid cavity; articulates with humeral head to form the glenohumeral joint

2.) Upper extremity



a.) Shoulder girdle

- Main bones: clavicle & scapula
- Function: connects the upper limb to the axial skeleton; provides mobility and muscle attachment

b.) Arm — brachium

- Main bones: humerus
- Function: single long bone of the arm; articulate with scapula at the shoulder and the radius and ulna at the elbow

c.) Forearm — antebrachium

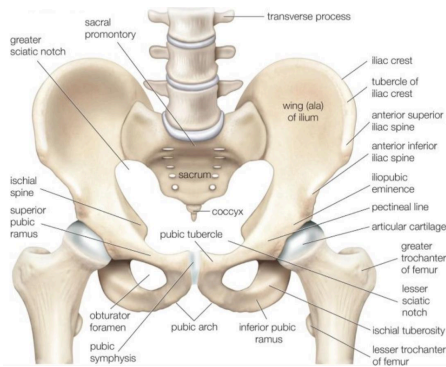
- Main bones: radius & ulna
- Function: forms forearm; allow pronation and supination (rotation of the hand)

d.) Hand — manus

- Main bones:
 - Carpals (8) - wrist bones
 - Metacarpals (5) - palm bones
 - Phalanges (14) - finger bones; no medial phalanx in thumb
- Function: most distal and functionally complex part; enables grasp, manipulation, and tactile sensation

3.) Pelvic girdle — hip girdle

WEEK 13 to 14: ORGAN SYSTEM



- Function:

- Weight bearing - supports the weight of the upper body when standing or sitting
- Weight transfer - Transfers body weight from the axial skeleton to the lower limbs during locomotion
- Protection - shields pelvic and lower abdominal organs (bladder, reproductive organs, and rectum)
- Attachment site - provides origin and insertion for many muscles involved in posture, locomotion, and limb movement

- **Bony ring**; connects the lower limbs to the axial skeleton

- Consist of 2 pelvic hip bones (os coxae) with each fused by 3 bones

- Ilium - superior, broad, fan-shaped portion
- Ischium - posteroinferior portion; bears weight when sitting
- Pubis - anteroinferior portion; contributes to pubic symphysis

- **Fusion**: 3 bones fuse at acetabulum, the deep socket for femoral articulation

- **Landmarks**: iliac crest, anterior superior iliac spine (ASIS), ischial tuberosity - distributes weight during sitting, pubic symphysis - muscle, ligament, and tendon attachment

- Articulations:

- **Sacroiliac joints — posterior**
 - Articulation between sacrum & ilium
 - Features: strong ligaments for stability; limited mobility
 - Function: transfers weight from the spine to the lower limbs
- **Hip joint — laterally**
 - Articulation between acetabulum of os coxae and the head of femur
 - Type: ball-and-socket synovial

joint

- Function: flexion, extension, abduction, adduction, rotation, and circumduction

4.) Lower extremity

- For support, weight-bearing, balance, and locomotion

- Consists of thigh, leg, and foot, with bones arranged to provide both stability and efficient movement

- More robust bones, stronger ligaments, and less mobility to facilitate standing and walking

a.) Thigh — femoral region

- Main bones: femur

- Function: Single bone; longest bone; articulates proximally with the acetabulum of the hip and distally with tibia and patella

b.) Leg — crural region

- Main bones: tibia (medial; weight-bearing), fibula (lateral; non-weight-bearing)

- Function: tibia transmits body weight from femur to the foot; fibula provide lateral stability and muscle attachment

c.) Foot — pedal region

- Main bones:

- Tarsals (7) - ankle bones; talus, largest in foot & heel bones; calcaneus, navicular, cuneiform (3), cuboid
- Metatarsals (5) - midfoot bones
- Phalanges (14) - toe bones; no medial phalanx in toe

- Function: form arches of the foot for shock absorption, balance, and propulsion during gait

d.) Patella

- Main bones: sesamoid bone embedded in quads tendon

- Function: Protects knee joint, increases leverage of quads during knee extension

FUNCTIONAL EMPHASIS

1.) Weight-bearing

- Transfers body weight from pelvis to ground through femur, tibia, and foot

- Bones are thicker and stronger

2.) Balance and stability

- Lower limb joints are more stable and less mobile than upper limb joints

WEEK 13 to 14: ORGAN SYSTEM

- Foot arches maintain balance and absorb impact during walking, running, and jumping

3.) Locomotion and propulsion

- Muscle and joints act in coordinated sequence for walking, running, and jumping

- Knee, ankle, and foot form a kinematic chain that efficiently transmits force to the ground

4.) Protection

- Tibia and fibula protect anterior and lateral leg structures; foot bones protect plantar structures

BIOMECHANICAL FEATURES

1.) Hip joint (acetabulofemoral) - ball-and-socket; high stability for weight-bearing, moderate mobility

2.) Knee joint - hinge-type; allows flexion, extension, and limited rotation; stabilized by ligaments (ACL, PCL, MCL, LCL)

3.) Ankle joint (talocrural) - hinge-type; dorsiflexion and plantarflexion

4.) Subtalar joint - plane-type; inversion and eversion of the foot

5.) Foot arches - longitudinal and transverse arches distribute body weight and provide spring during gait

